

Krishnaa Vadivel | PhD Research Engineer | Computational Physics | EE Systems | Scientific ML

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Profile

Broad-profile electrical engineering PhD researcher with training and project work spanning computational physics, semiconductor and photonics theory, scientific machine learning, embedded systems, digital design, and high-performance scientific software. The core strength is connecting mathematical models, physical intuition, and implementation detail across research and engineering settings.

Research Experience

Yale University, Electrical Engineering

Research Assistant / PhD Researcher

2021–present

- Develop first-principles and many-body approaches for self-trapped excitons, excitonic polarons, and electron-phonon coupled systems, connecting DFT, DFPT, excited-state theory, and effective Hamiltonian viewpoints.
- Study coherent-state and path-integral formulations for interacting quantum systems, including time-dependent and self-localized phenomena relevant to condensed-matter and materials problems.
- Build distributed scientific-computing workflows in Python and C++ using MPI, OpenMP, PETSc, and SLEPc for large simulation and linear-algebra workloads.
- Port and optimize workloads for accelerator-enabled HPC systems with CUDA and ROCm, with practical experience on Frontier, Perlmutter, Frontera, and related clusters.
- Explore ML accelerators for scientific computing, including equivariant graph neural networks for molecular-property prediction and agentic retrieval workflows for generating structured computational-science inputs.
- Support embedded-systems instruction and hardware-oriented lab work through teaching-assistant responsibilities tied to firmware, debugging, and instrumentation.
- Co-authored a 2025 *Journal of the American Chemical Society* paper on ultrafast self-trapped exciton formation in lead-free double halide perovskites and presented related APS talks in 2024, 2025, and 2026.

Professional Experience

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Built CUDA-based data-processing pipelines and custom OpenGL visualization software for high-volume acoustic waveform datasets used in deployed well-logging tools.
- Developed embedded software and digital logic for sensing, logging, and control functions using Lattice FPGA, PIC32, dsPIC, STM32, and C8051-based systems.
- Supported board- and system-level engineering tasks including oscilloscope-based debugging, integration testing, field validation, and selective PCB modifications in Altium.
- Contributed host-side tooling with C#, ASP-style services, and Microsoft SQL-backed database workflows supporting engineering operations and data handling.

FindLight

Photonics Technical Writer

2018–2019

- Produced technical articles and engineering-facing content covering lasers, optics, photonics components, and optoelectronic systems.

Selected Projects

Distributed Exciton-Phonon and Materials Simulation

- Developed scalable first-principles and many-body workflows for excited-state materials problems, emphasizing reusable HPC infrastructure and theory-to-code translation.

Semiconductor Device Simulation

- Built numerical treatments of electrostatics and transport equations, including Poisson, drift-diffusion, and pn-junction-style models linking semiconductor theory with simulation practice.

Equivariant Molecular Machine Learning

- Implemented graph-based and symmetry-aware ML models for molecular-property prediction and investigated their role in accelerating scientific workflows.

FPGA AI Accelerator and Digital Design Work

- Implemented Verilog-based accelerator and processor projects to demonstrate datapath design, verification workflows, and hardware-oriented compute thinking.

Embedded Robotics and Human-Interface Projects

- Built robotics and embedded-system class projects including an EEG-driven robotic hand and maze-navigation work with wireless coordination.

Integrated Photonics and FDTD Modeling

- Simulated band-pass and resonator-style photonic structures with Meep-based FDTD workflows for integrated-optics exploration.

Agentic Scientific Automation

- Built retrieval-driven tooling that reads software documentation and research papers to generate structured inputs and workflow scaffolds for computational science.

Education

Yale University

PhD, Electrical Engineering

2021–present

Ongoing work spanning computational materials, many-body theory, semiconductor-relevant modeling, and scientific software.

Yale University

MPhil, Electrical Engineering

2023

Yale University

MS, Electrical Engineering

2023

Cornell University

MEng, Electrical and Computer Engineering

2017–2018

Included stochastic-computing-related work, device-oriented EE topics, and advanced design coursework.

Cornell University

BS, Electrical and Computer Engineering

2013–2017

Included embedded systems, robotics, controls, and strong math/physics/computing preparation.

Selected Publications

2025: de Paula, A. M., Li, S., Hou, B., Vadivel, S., Teles-Ferreira, D. C., Iudica, A., Kabacinski, P., Hosseini, H., McArthur, J., Cerullo, G., et al. (2025). Time-domain observation of ultrafast self-trapped exciton formation in lead-free double halide perovskites. *Journal of the American Chemical Society*, 147(32), 28923–28931.

Selected Conference Presentations

2026: Vadivel, S., Grande, R. D., Strubbe, D. A., & Qiu, D. Y. (2026). First-principles study of exciton-polarons and self-trapped excitons. *APS Global Physics Summit 2026*.

2025: Vadivel, S., Qiu, D. Y., Grande, R. D., & Strubbe, D. A. (2025). First-principles study of self-trapped exciton formation in double perovskites. *APS Global Physics Summit 2025*.

2024: Vadivel, S., & Qiu, D. (2024). First-principles study of self-trapped exciton formation in double perovskites using excited state forces. *APS March Meeting Abstracts 2024*, F01–007.

Selected Coursework

Mathematics: Differential Geometry; Abstract Algebra; Honors Mathematical Analysis II

Computer Science: Theoretical Computer Science; Probabilistic Machine Learning

Physics: Graduate Quantum Mechanics I–II; Solid State Physics I–II; Many-Body Quantum Physics

Electrical Engineering: Semiconductor Physics; Thin Film Science; Lasers and Optoelectronics; Integrated Photonics; VLSI Design; Analog VLSI Design; Mathematics of Signals and Systems; Advanced Embedded Systems

Technical Skills

Languages: Python, C, C++, CUDA, JavaScript, Bash, PowerShell, C#

Scientific Computing: MPI, OpenMP, PETSc, SLEPc, mpi4py, petsc4py, slepc4py, NumPy, pandas, SciPy, SymPy, matplotlib

Machine Learning: PyTorch, PyTorch Geometric, scikit-learn, equivariant graph neural networks, scientific ML workflows

Hardware: Verilog, Lattice FPGA, PIC32, dsPIC, STM32, C8051, sensor interfaces, PCB bring-up, oscilloscope debugging

Software / Systems: OpenGL, ASP-style services, Microsoft SQL, Linux command line, Docker, Docker Compose, Kubernetes, gcloud

HPC Platforms: Frontier, Frontera, Perlmutter, and related national-lab or university HPC environments